PROMPT LIFE V0.18.0

DiagramKit + Illustration Clarity Foundation

Reusable coded diagram primitives, four Before Morning diagram refactors, review-gallery coverage, mobile screenshots, and verification results.

1. What Changed

- Added a local DiagramKit component module for reusable SVG/HTML teaching diagram primitives.
- Refactored four coded Before Morning visuals: `ai-family-tree`, `traditions`, `training-loop`, and `overfitting-generalization`.
- Added a DiagramKit primitive gallery to `/review/visual-aids`.
- Bumped the visible app version to `0.18.0`.
- Added visual grammar, refactor audit, tiny-interaction triage, review notes, screenshots, and this PDF report.

2. Files Changed

- `src/components/DiagramKit.tsx`
- `src/components/VisualAids.tsx`
- `src/styles/global.css`
- `src/main.tsx`
- `docs/design/DIAGRAM_KIT_V0_18.md`
- `docs/curriculum/TINY_INTERACTION_TRIAGE_V0_18.md`
- `docs/stage-audits/v0-17-before-morning/DIAGRAM_KIT_REFACTOR_V0_18.md`
- `docs/REVIEW_NOTES.md`
- `docs/stage-audits/v0-17-before-morning/screenshot-index.md`
- `docs/stage-audits/v0-17-before-morning/screenshots/v0-18-*.png`

3. Library / Dependency Decisions

No new dependencies were added. The visuals use local React/SVG primitives rather than D3, icon packages, generated PNGs, or 3D libraries because the required diagrams are sparse, mobile-first teaching scenes.

4. DiagramKit Components Created

`DiagramFrame`, `DiagramScene`, `PaperNode`, `PaperPanel`, `NeonPath`, `NeonArrow`, `CalloutSeal`, `TokenChip`, `ContextTray`, `VectorBar`, `ProbabilityBar`, `LayerSheet`, `TensorGrid`, `EvidenceCard`, `GuardrailPath`, `WarningZone`, `GlowNode`, `LegendRow`, `DiagramCaption`, and `DiagramCallouts`.

5. Visuals Refactored

VISUAL	BEHAVIOR
Where LLMs Fit	Folded AI family tree with short category labels and HTML callouts for the details.
Rationalists vs Empiricists	Two paper panels plus a bridge showing rules-first and learned-pattern systems can combine.
Training	Five-step loop with `Update weights` highlighted as the durable-change step.
Overfitting and Generalization	Readable plot with training dots, held-out examples, overfit curve, smoother curve, and compact legend.

6. Tiny Interaction Triage

The pass intentionally did not tune interaction mechanics. The triage recommends future focused work on Training, Rationalists vs Empiricists, and Overfitting while preserving answer randomization, progress storage, non-competitive tone, and the single badge model.

7. Build / Typecheck / Audit Result

- `npm run typecheck`: passed.
- `npm run build`: passed with the existing Vite large-chunk warning.
- `npm run build:pages`: passed with the existing Vite large-chunk warning.
- `npm run audit:answers`: passed; 68 surfaces, 46 randomized, 22 fixed-order exclusions.
- Browser QA at 390px and 320px: screenshots captured; no console errors found.
- Smoke checks: main app, lesson review route, visual-aid review route, and Badge version render correctly.

8. Bundle-Size Impact

ASSET	BASELINE	V0.18	APPROX. CHANGE
CSS raw	68.76 kB	73.47 kB	+4.71 kB
CSS gzip	15.00 kB	15.97 kB	+0.97 kB
JS raw	630.81 kB	638.93 kB	+8.12 kB
JS gzip	170.95 kB	173.88 kB	+2.93 kB

9. Known Issues

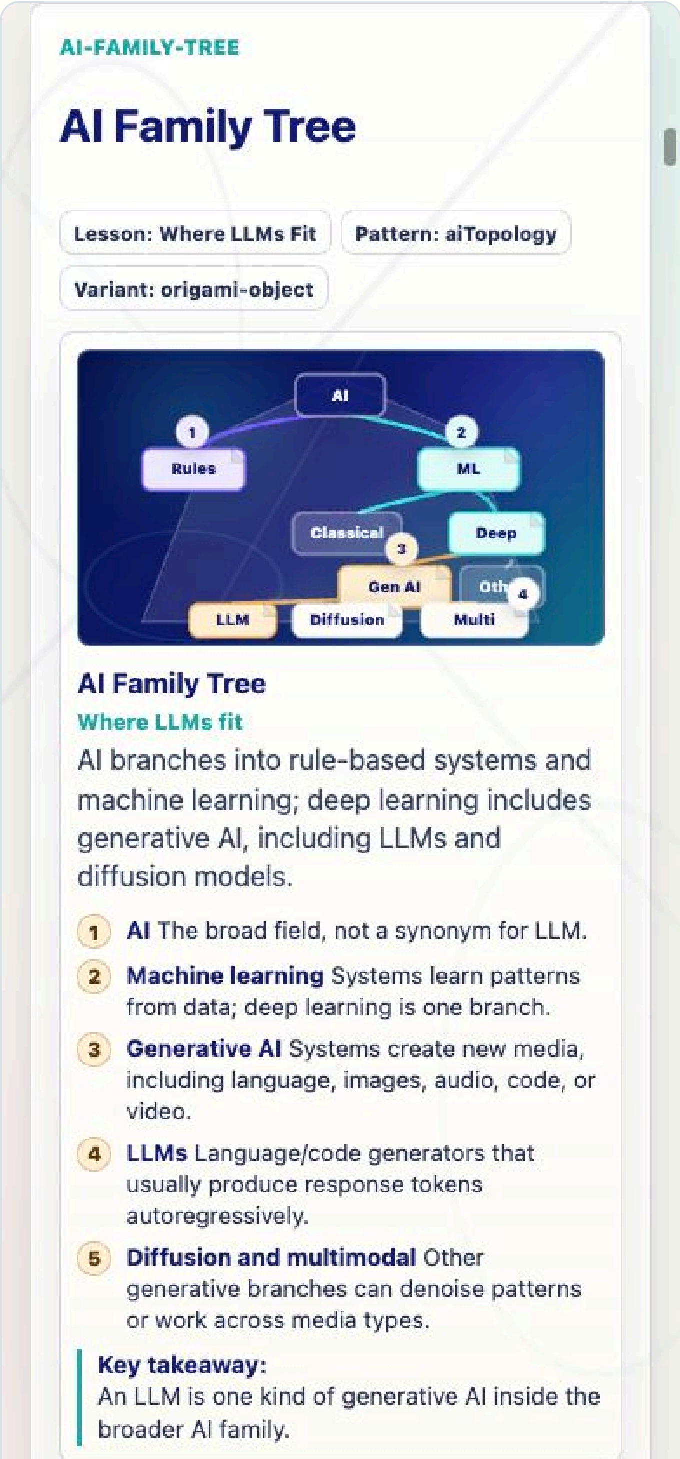
- The existing Vite large-chunk warning remains.
- DiagramKit is a focused teaching-diagram toolkit, not a general charting library.
- Generated Image 2 assets should later be reviewed against this visual grammar so coded and image-backed visuals continue to feel related.

10. Next Three Recommended Steps

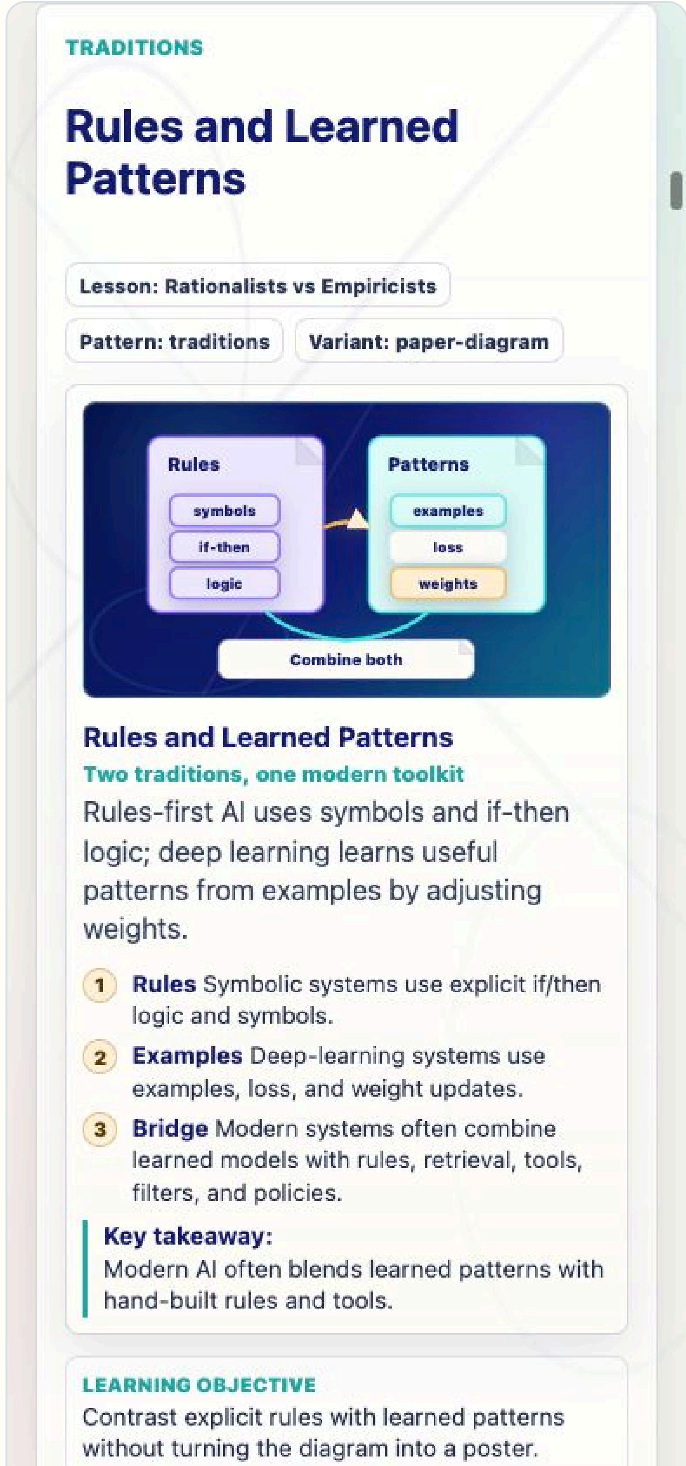
1. Generate the next Before Morning Image 2 asset only after confirming which remaining coded visuals should stay coded.
2. Run a small tiny-interaction clarity pass on Training, Traditions, and Overfitting.
3. Plan app-level code splitting to retire the persistent Vite large-chunk warning.

Image 2 readiness: Before Morning is ready for Image 2 asset generation from a visual-system standpoint.

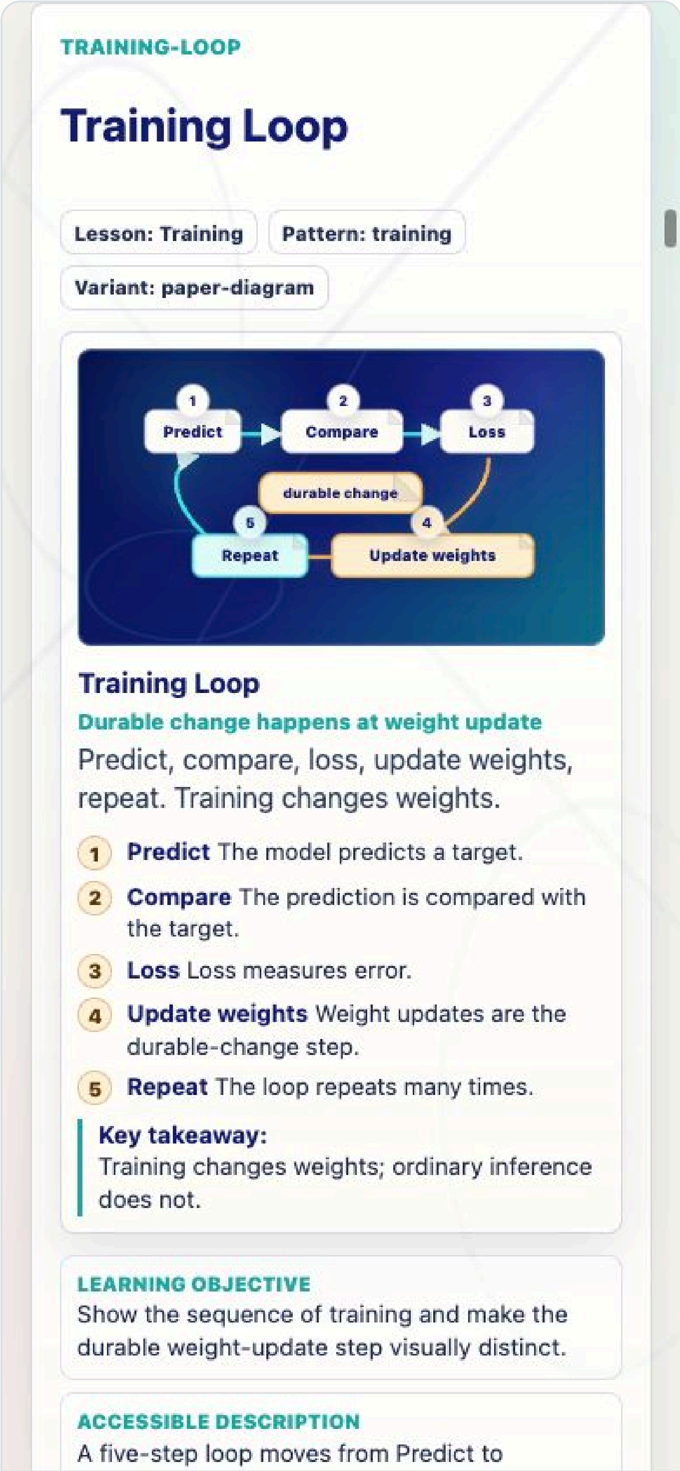
Screenshots



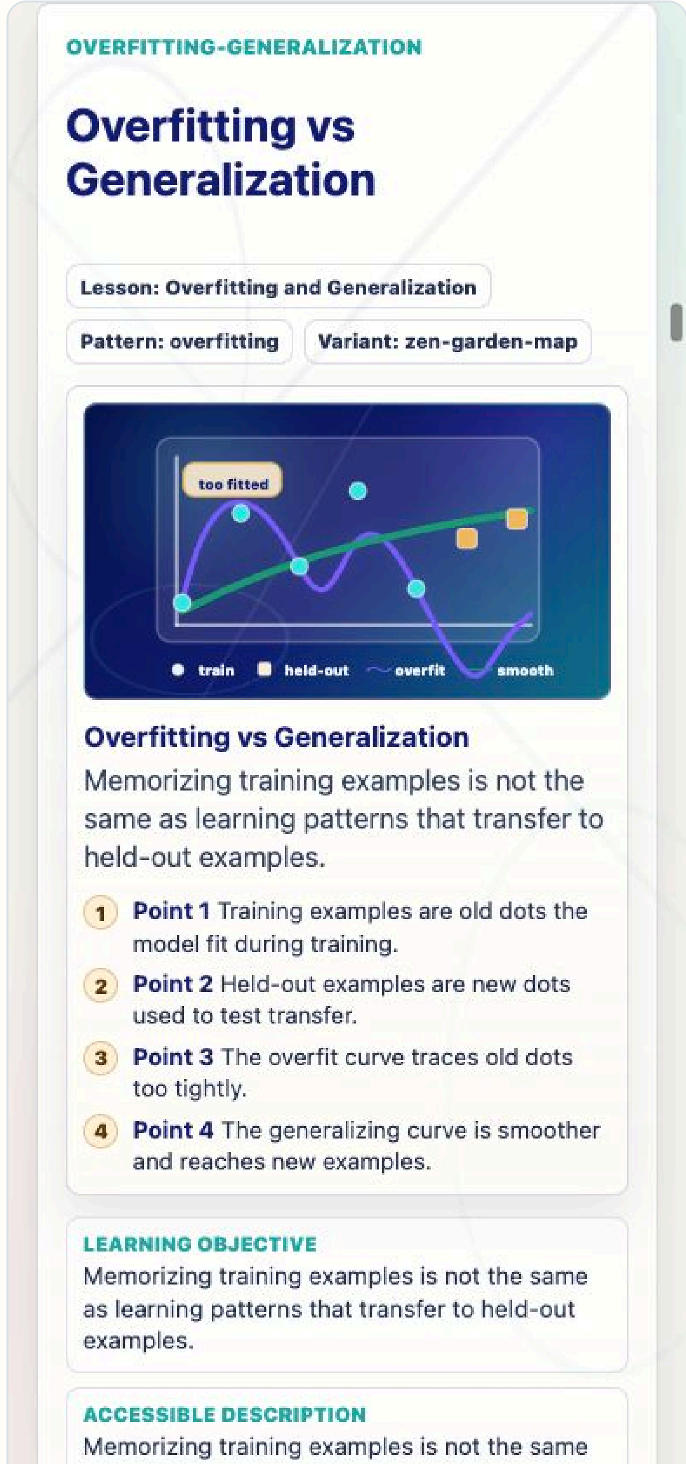
Where LLMs Fit: AI family tree at 390px.



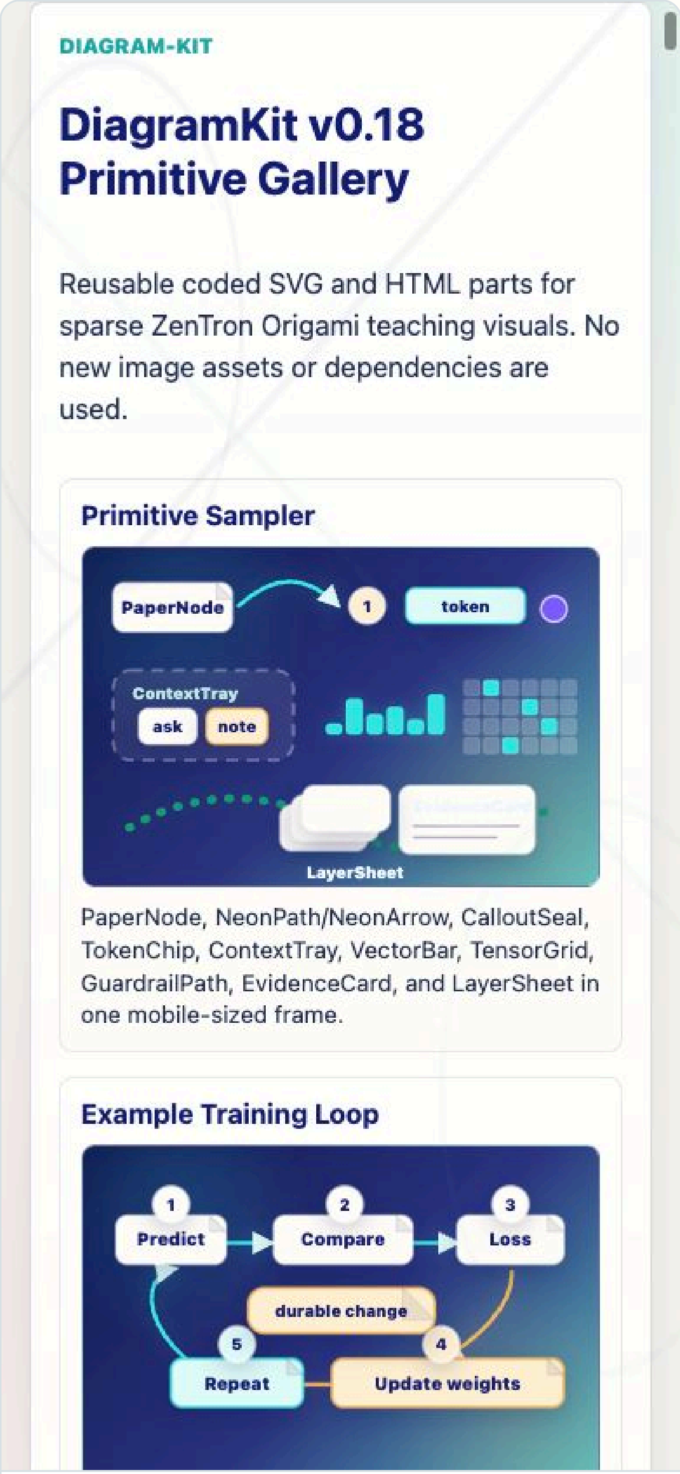
Rationalists vs Empiricists at 390px.



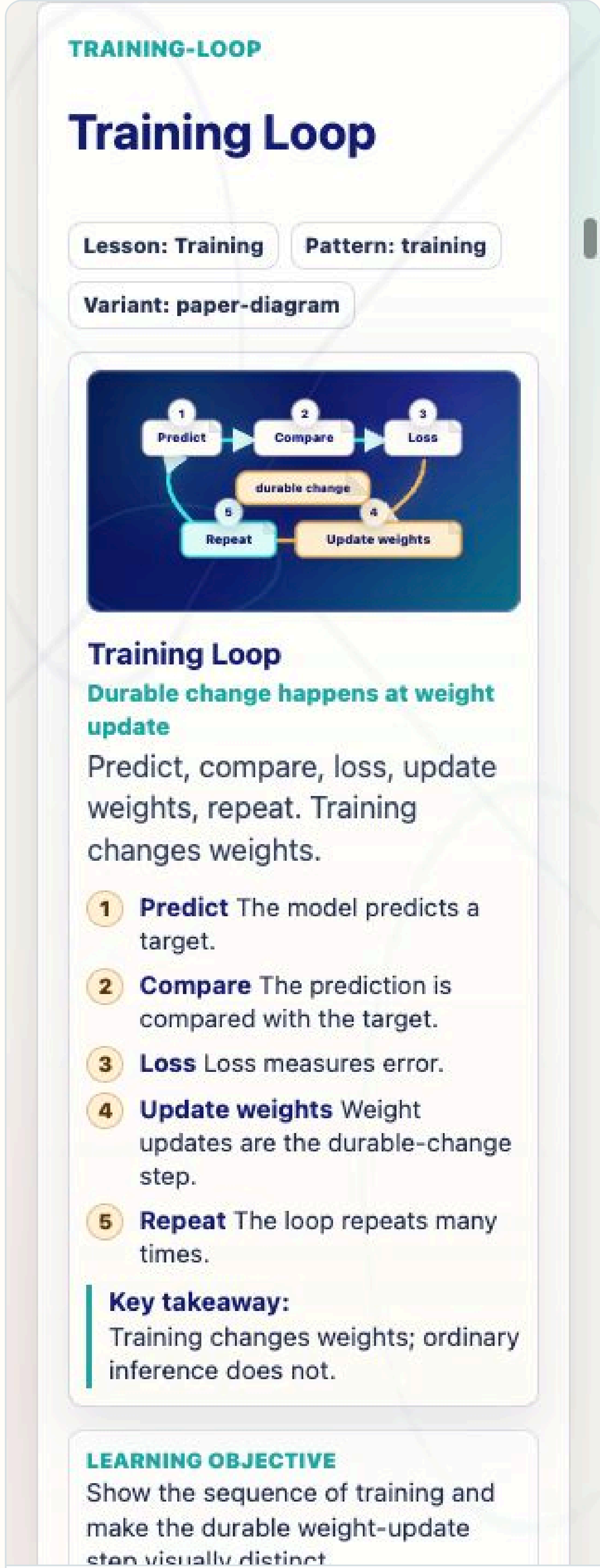
Training loop at 390px.



Overfitting and Generalization at 390px.



DiagramKit gallery at 390px.



Training loop at 320px.

BEFORE-MORNING-LLM-CLOUD

Prompt to Prediction

Lesson: What Is an LLM?

Pattern: generatedImage

Variant: generated-image

Asset: before-morning-llm-cloud.png

Type: generated-image

Prompt to Prediction

Generated asset still working: Prompt to Prediction at 390px.

A textless model-cloud scene is paired with HTML callouts for context, learned weights, one generated token, and append-repeat.

1. **Context enters** The current prompt and response-so-far become the visible input for this run.
2. **Weights shape probabilities** Learned weights, built earlier during training, shape next-token scores.
3. **One token is generated** The model chooses one response token from the probability cloud